

Retail Sales Analysis Using Excel

1. Project Overview

This project simulates a real-world scenario where a marketing analyst uses Excel to evaluate customer purchase data. The goal is to extract insights using Conditional Formatting and essential Excel functions to drive data-based decision-making.

2. Tasks Performed (with optional screenshots)

✓ Task 1: Sales Performance Highlight

- **Goal:** Highlight revenue > \$500
- **Method:** Used Conditional Formatting with a numeric rule
- **Formula/Steps:**
 - Go to Conditional Formatting → Highlight Cells → Greater Than → 500

CustomerID	ProductCategory	PurchaseDate	Revenue	Age	Region	PurchaseFrequency
78268	Clothing	20-09-2023	917.320545	55	North	4
53567	Sports	21-07-2023	194.004026	56	North	3
52613	Books	01-05-2023	355.087898	46	North	1
55891	Clothing	10-03-2023	265.212104	45	South	5
31243	Books	17-03-2023	161.836644	63	West	3
51993	Home Decor	10-02-2023	898.105374	51	South	1
96293	Home Decor	18-07-2023	933.118995	29	West	9
62026	Clothing	13-09-2023	999.215829	35	North	2
90471	Electronics	08-07-2023	244.385938	52	North	1
56006	Electronics	04-03-2023	156.889383	31	West	2
49512	Clothing	06-02-2023	730.235821	33	North	4
62620	Clothing	18-03-2023	930.381359	42	South	2
90186	Sports	01-04-2023	293.510855	30	East	3
27089	Electronics	28-06-2023	812.36725	56	East	4
42230	Electronics	01-08-2023	798.820741	62	South	1
28983	Home Decor	25-07-2023	957.957701	64	South	6
99688	Books	21-04-2023	458.433264	53	West	2
92457	Home Decor	13-06-2023	644.956643	19	South	1
16921	Electronics	03-05-2023	452.081747	60	South	3
48004	Sports	30-05-2023	457.406327	45	North	2

✓ Task 2: Monthly Sales Trends

- **Goal:** Highlight revenue values equal to the average
- **Method:** Used AVERAGE function in a custom rule
- **Formula:**

=D2=AVERAGE(\$D\$2:\$D\$1001)

CustomerID	ProductCategory	PurchaseDate	Revenue	Age	Region	PurchaseFrequency
78268	Clothing	20-09-2023	917.320545	55	North	4
53567	Sports	21-07-2023	194.004026	56	North	3
52613	Books	01-05-2023	355.087898	46	North	1
55891	Clothing	10-03-2023	265.212104	45	South	5
31243	Books	17-03-2023	161.836644	63	West	3
51993	Home Decor	10-02-2023	898.105374	51	South	1
96293	Home Decor	18-07-2023	933.118995	29	West	9
62026	Clothing	13-09-2023	999.215829	35	North	2
90471	Electronics	08-07-2023	244.385938	52	North	1
56006	Electronics	04-03-2023	156.889383	31	West	2
49512	Clothing	06-02-2023	730.235821	33	North	4
62620	Clothing	18-03-2023	930.381359	42	South	2
90186	Sports	01-04-2023	293.510855	30	East	3
27089	Electronics	28-06-2023	812.36725	56	East	4
42230	Electronics	01-08-2023	798.820741	62	South	1
28983	Home Decor	25-07-2023	957.957701	64	South	6
99688	Books	21-04-2023	458.433264	53	West	2
92457	Home Decor	13-06-2023	644.956643	19	South	1
16921	Electronics	03-05-2023	452.081747	60	South	3
48004	Sports	30-05-2023	457.406327	45	North	2

✓ Task 3: Region-wise Revenue Summary

- **Goal:** Summarize revenue by region
- **Method:** Used SUMIFS function
- **Formula:**

=SUMIFS(Revenue, Region, "North")

CustomerID	ProductCategory	PurchaseDate	Revenue	Age	Region	PurchaseFrequency					
78268	Clothing	20-09-2023	917.320545	55	North	4					
53567	Sports	21-07-2023	194.004026	56	North	3					
52613	Books	01-05-2023	355.087898	46	North	1					
55891	Clothing	10-03-2023	265.212104	45	South	5			North	3897.656	
31243	Books	17-03-2023	161.836644	63	West	3			South	4947.516	
51993	Home Decor	10-02-2023	898.105374	51	South	1			East	1105.878	
96293	Home Decor	18-07-2023	933.118995	29	West	9			West	1710.278	
62026	Clothing	13-09-2023	999.215829	35	North	2					
90471	Electronics	08-07-2023	244.385938	52	North	1					
56006	Electronics	04-03-2023	156.889383	31	West	2					
49512	Clothing	06-02-2023	730.235821	33	North	4					
62620	Clothing	18-03-2023	930.381359	42	South	2					
90186	Sports	01-04-2023	293.510855	30	East	3					
27089	Electronics	28-06-2023	812.36725	56	East	4					
42230	Electronics	01-08-2023	798.820741	62	South	1					
28983	Home Decor	25-07-2023	957.957701	64	South	6					
99688	Books	21-04-2023	458.433264	53	West	2					
92457	Home Decor	13-06-2023	644.956643	19	South	1					
16921	Electronics	03-05-2023	452.081747	60	South	3					
48004	Sports	30-05-2023	457.406327	45	North	2					

3. Insights and Findings

- ✦ Conditional Formatting clearly revealed high-performing sales.
 - ✦ No values matched the exact average, indicating skewed or non-uniform data.
 - ✦ Region-wise summaries made it easy to identify top-performing markets.
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4. Tools Used

- Microsoft Excel
 - Conditional Formatting
 - Excel Functions: AVERAGE, SUMIFS
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5. Conclusion

This project enhanced my ability to analyse data using Excel's advanced tools. I learned how to format and summarize information in a way that improves readability, insight generation, and communication of results — critical skills for any analyst.

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